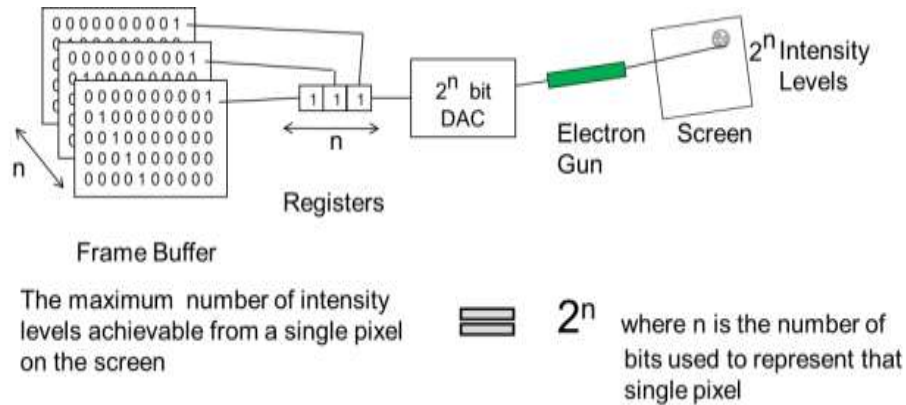


Digital to Analogue Converter (DAC)

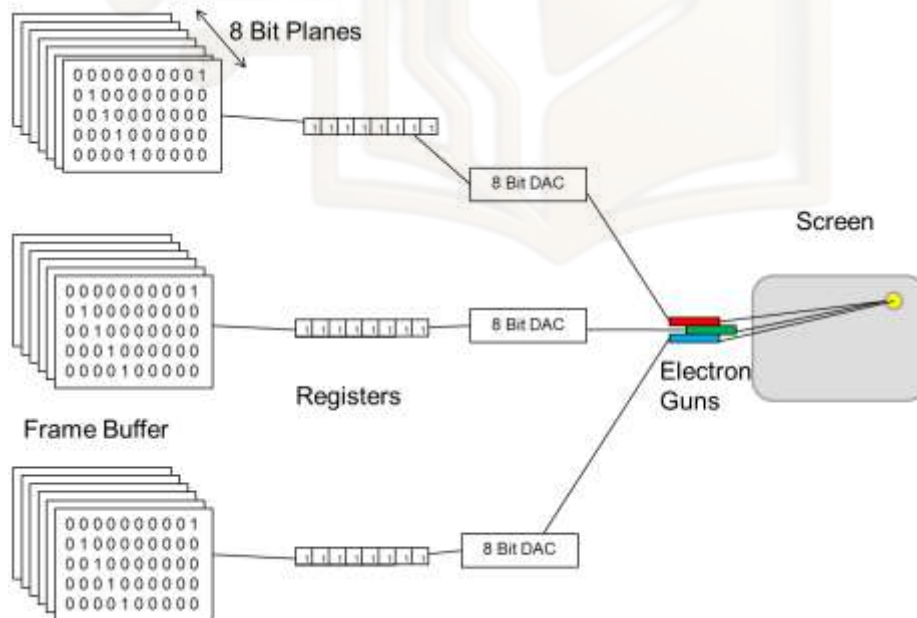
It converts a digital input signal into an analog output signal

In case of a Raster display system, the frame buffer is a digital device and the screen is an analogue device. For display generating purpose, the digital to analogue converter accesses the bits in the frame buffers and converts it into analogue signal necessary to signal the electron gun or other components to glow the pixels on the screen.



An 'n' bit DAC can generate 2ⁿ voltage signals for the electron gun to excite the phosphors dots on the screen thus generating 2ⁿ shades of that color (red, green or blue) on the screen.

So if an 8 bit DAC is used for each of the three components (red, green or blue) then a total number of 24 bits are used to represent a single pixel on the screen comprised of red, green and blue phosphor dots which means that the DAC in total can produce 2²⁴ = 16777216 different voltage levels for the electron guns thus producing 16777216 different intensities out of a single pixel on the screen.



Frame Buffer Organization

It is the contiguous memory allocation that is used to hold pixel intensity values of the image to be displayed on a computer screen.

It may be a part of system memory or a part of separate memory on the graphics cards.

The size of the frame buffer depends upon the number of bits allocated per pixel.

So if the total resolution of the screen is 640 x 480 and each pixel is allocated 1 single bit.

Then the total size of the frame buffer is 640 x 480 = 307200 bits.

If the total resolution of the screen is 640 x 480 and each pixel comprised of red, green and blue components represented by 8 bits for each components then each pixel is represented by 24 bits
Then the total size of the frame buffer is 640 x 480 x 12 bits

Video Controller Refresh Operation

A Video controller reads the intensity information for each pixel stored in the frame buffer and controls the intensity of the pixel on the screen

